

AMAZON INDIA

SALES PERFORMANCE ANALYSIS

April 2022 – June 2022 | 128,975 Orders | Tools: MySQL · Power BI · Python

Data Analytics Portfolio Project

KEY PERFORMANCE METRICS

1,28,975 Total Orders Apr–Jun 2022	₹7.34 Cr Total Revenue Gross Sales Value	₹609.83 Avg Order Value Per Transaction	13.32% Cancellation Rate 17,185 orders lost
65.1% Amazon Fulfilled 84,002 orders	68.9% Expedited Shipping 82,922 orders	SET Top Category ₹3.67 Cr revenue	Maharashtra Top State ₹1.25 Cr revenue

1. Project Overview

This project presents a comprehensive analysis of Amazon India's fashion e-commerce sales data covering April–June 2022. The objective was to extract actionable business insights from raw transactional data using SQL for data querying, Power BI for interactive visualization, and Python (Pandas) for data validation and profiling.

Objectives

- Understand overall revenue performance, order volume, and customer behaviour
- Identify top-performing product categories, cities, and states
- Evaluate operational metrics: fulfilment types, shipping modes, and cancellation rates
- Segment states by revenue tier for strategic planning
- Build a recruiter-ready portfolio demonstrating end-to-end data analytics skills

Tools & Technologies Used

Tool	Purpose
PostgreSQL	Data extraction, aggregation, window functions, CTEs (17 analytical queries)
Power BI	Interactive dashboard with slicers, KPI cards, bar/pie charts, map visuals
Excel	Data Cleaning and Tranformation
Dataset	128,975 rows × 21 usable columns — Amazon India fashion orders (Apr–Jun 2022)

2. Dataset Overview

The dataset contains real-world Amazon India transactional records for fashion merchandise. Each row represents a single order with attributes covering order identity, product details, shipping information, fulfilment type, and payment amount.

Field Group	Key Columns	Description
Order Identity	Order ID, Date, Status	Unique order reference, order date, fulfilment status
Product Details	Category, SKU, Style, Size	Fashion product classification and size information
Shipping Info	ship-city, ship-state, service level	Delivery location and shipping method
Financials	Amount, Qty	Order value in INR and units sold
Fulfilment	Fulfilment, B2B, fulfilled-by	Amazon vs Merchant fulfilled; B2B vs B2C orders

3. SQL Analysis & Key Findings

Seventeen SQL queries were developed to answer targeted business questions. Queries spanned basic aggregations, GROUP BY analysis, CASE-WHEN logic, Common Table Expressions (CTEs), and window functions (NTILE, RANK). Below are the major findings organised by theme.

3.1 Revenue & Order Volume

Metric	Value
Total Orders	1,28,975
Total Revenue	₹7,34,10,501 (~₹7.34 Crore)
Average Order Value	₹609.83
Best Revenue Month	April 2022 — ₹2.70 Crore (36.7% share)

Revenue showed a clear downward trend month-over-month: April (₹2.70 Cr) → May (₹2.45 Cr) → June (₹2.19 Cr). This decline of ~18.8% over the period warrants investigation into seasonal demand, marketing spend, or inventory constraints.

3.2 Product Category Performance

The fashion catalogue was divided into 9 categories. 'Set' and 'Kurta' together accounted for 86.4% of total revenue, establishing them as the core revenue drivers.

Category	Revenue (₹)	Orders	Cancellation %
Set	3,67,31,224	47,207	14.60%
Kurta	1,96,00,206	45,917	14.66%
Western Dress	1,06,28,975	14,713	13.76%
Top	50,30,323	10,021	12.13%
Ethnic Dress	7,32,526	1,081	12.49%
Others (Blouse, Bottom, Saree, Dupatta)	6,87,248	1,439	~13%

Insight: 'Set' was the #1 category every single month (Apr–Jun). Despite high volume, Kurta and Set have the highest cancellation rates (~14.6%), suggesting a potential sizing, quality-expectation, or returns-policy issue worth addressing.

3.3 Geographic Distribution

Order delivery was spread across all major Indian states. The top 5 states collectively drove ~55% of total revenue.

Rank	State	Revenue (₹)	Tier
1	Maharashtra	1,24,55,530	Top Tier
2	Karnataka	98,29,397	Top Tier
3	Uttar Pradesh	64,40,187	Top Tier
4	Telangana	63,38,385	Top Tier
5	Tamil Nadu	59,69,968	Top Tier

Top 5 cities by shipped orders: Bengaluru (7,262) › Hyderabad (5,271) › Mumbai (4,158) › Chennai (3,691) › New Delhi (3,660). Metro cities dominate fulfilment volume, reflecting strong urban demand for fashion e-commerce.

3.4 Fulfilment & Shipping Analysis

Dimension	Segment	Orders	Revenue (₹)
Fulfilment Type	Amazon	84,002 (65.1%)	5,08,86,049
Fulfilment Type	Merchant	36,376 (28.2%)	2,25,24,452
Shipping Mode	Expedited	82,922 (68.9%)	5,08,50,328
Shipping Mode	Standard	37,456 (31.1%)	2,25,60,173

Amazon Fulfilled orders command a significantly higher revenue share (69.3%) compared to Merchant Fulfilled (30.7%), indicating higher basket sizes or premium product listings on FBA. Expedited shipping dominates, reflecting customer preference for speed — a potential upsell opportunity for Amazon Prime promotions.

3.5 Order Status & Cancellation Analysis

Order Status	Count	% Share
Shipped	72,826	56.5%
Shipped – Delivered to Buyer	26,566	20.6%
Cancelled	17,185	13.32%
Shipped – Returned to Seller	1,851	1.44%
Pending / Other	854	0.66%

A cancellation rate of 13.32% is above the typical 5–8% benchmark for fashion e-commerce. Saree (14.69%), Kurta (14.66%), and Set (14.60%) are the worst-performing categories by cancellation. This signals potential product-listing issues, unrealistic delivery promises, or a cash-on-delivery preference with high buyer drop-off.

3.6 B2B vs B2C Segmentation

The dataset revealed a heavily B2C-dominated sales structure:

- B2C orders: 1,19,584 (98.8% of all orders) — retail consumers
- B2B orders: 794 (0.62% of all orders) — business buyers

While B2B volume is low, business customers often place larger-value bulk orders. Expanding the B2B segment with dedicated account management could represent a meaningful revenue diversification opportunity.

3.7 Advanced SQL — Window Functions & CTEs

Beyond basic queries, three advanced techniques were applied to extract deeper insights:

Technique	Query	Insight Derived
NTILE(3) Window Function	Query 16	States classified as Top / Mid / Low Tier by revenue, enabling tiered marketing budgets
RANK() with PARTITION BY	Query 17	Top performing category identified for each month — 'Set' ranked #1 in all three months
CTE + Subquery	Query 16	Modular CTE improves query readability and enables reuse in downstream reporting

4. Power BI Dashboard

An interactive Power BI dashboard (.pbix) was built to visually communicate the SQL findings. The dashboard enables non-technical stakeholders to explore data through dynamic filters and drill-down capabilities.

Dashboard Components

Visual Type	Business Question Answered
KPI Cards (4)	Total Orders, Revenue, AOV, Cancellation Rate at a glance
Clustered Bar Chart	Revenue & order count by Category — instant identification of top/bottom performers
Donut Chart	Amazon vs Merchant fulfilment split; Expedited vs Standard shipping
Map Visual	Geographic revenue distribution across Indian states
Line / Column Chart	Month-over-month revenue trend (Apr → May → Jun 2022)
Slicers	Dynamic filtering by Month, Category, State, Fulfilment Type, Shipping Mode

5. Business Insights & Recommendations

The following actionable recommendations emerge directly from the data analysis. Each is tied to a specific SQL finding.

💡 Insight 1 — Revenue Decline Needs Immediate Attention

Revenue dropped 18.8% from April to June. A detailed cohort analysis of new vs returning customers per month is recommended to determine whether the drop is driven by acquisition slowdown or reduced repeat purchase rates.

💡 Insight 2 — Reduce Category-Level Cancellations

Kurta and Set have cancellation rates above 14.5%, significantly above platform average. Actions: improve product photography, add accurate size guides, introduce Try-Before-You-Buy for high-value sets, and analyse COD vs prepaid cancellation split.

💡 Insight 3 — Double Down on Top-Tier States

Maharashtra and Karnataka alone account for 30.1% of revenue. Targeted marketing campaigns, exclusive launches, and same-day delivery pilots in Mumbai, Pune, Bengaluru, and Hyderabad would maximise ROI on existing demand.

💡 Insight 4 — Expand B2B Channel

With only 0.62% B2B share, there is substantial untapped potential in bulk/business orders for uniforms, gifting, and resale. A dedicated B2B portal or account manager programme for boutique resellers could significantly expand this segment.

💡 Insight 5 — Upsell Expedited Shipping

68.9% of orders already use expedited shipping, and these generate 69.2% of revenue. Converting the remaining 31.1% standard-shipping customers through Amazon Prime promotions or conditional free-expedited-shipping offers could meaningfully boost average order value.

6. Skills Demonstrated

This project was designed to showcase a full-stack data analytics skill set relevant to entry-to-mid level Data Analyst roles.

Skill Area	Demonstrated Through
SQL (Intermediate–Advanced)	17 queries covering aggregations, GROUP BY, HAVING, CASE-WHEN, CTEs, NTILE, RANK, PARTITION BY
Data Analysis	Trend analysis, segmentation, cancellation analysis, fulfilment benchmarking, revenue tiering
Data Visualisation	Power BI dashboard with KPI cards, map, bar, donut, line charts and interactive slicers
Excel	Dataset cleaning and Transformation
Business Acumen	Translating raw data findings into actionable business recommendations with revenue impact context
Communication	Structured written report with executive summary, insight narratives, and recommendation framing

7. Conclusion

This project successfully demonstrates the ability to take a raw, real-world e-commerce dataset and transform it into structured, decision-ready business intelligence. Through 17 SQL queries, an interactive Power BI dashboard, and Python-based data profiling, this analysis uncovered:

- A ₹7.34 Crore revenue base with clear category and geographic concentration
- A 13.32% cancellation rate representing a significant revenue recovery opportunity
- Amazon Fulfilment and Expedited Shipping as the dominant and higher-value operational modes
- 'Set' as the consistently top-performing category across all three months
- Maharashtra and Karnataka as the highest-priority states for marketing investment

The structured approach — from data ingestion and SQL querying, through Power BI visualisation, to business recommendation framing — mirrors the end-to-end workflow expected of a professional Data Analyst. This project is part of a growing portfolio available on GitHub and LinkedIn.

— End of Report —